

Chapter 1

Supersonic Flow - Physical and Modeling Aspects

“The importance of the Mach number lies in the fact that it is a measure of the ratio of the directed kinetic energy of the flow to the random thermal energy of the molecules.”

— John D. Anderson, Jr.

PREVIEW

In the preceding chapters of this textbook, we navigated the elegant simplicity of incompressible aerodynamics, where density was a well-behaved constant. In this chapter, we shatter that assumption. We move into the energetic world of high-speed flow where density becomes a dynamic participant, temperature rises sharply across infinitesimally thin discontinuities, and the fluid’s communication system fundamentally breaks down. We will explore the formation of shock waves, derive the Rankine–Hugoniot relations, examine the phenomenon of shock detachment, and demonstrate how a modern density-based CFD solver such as ANSYS Fluent bridges the gap between abstract theory and engineering reality.

ROADMAP

We begin with the engineering motivation for studying compressible flow, then build the physical intuition of compressibility and the propagation of acoustic disturbances. We next reduce the full Navier–Stokes system to the Euler equations and justify this simplification. A rigorous derivation of normal and oblique shock relations follows, accompanied by a discussion of expansion fans and the distinction between attached and detached shocks. We then transition to the digital laboratory, detailing the geometry, mesh, boundary conditions and solver settings used in ANSYS Fluent. Finally, we analyse the simulation results for $M_\infty = 2, 3$ and 4, compare them against theory, and critically assess the limitations of the inviscid model.

1.1 Introduction and Engineering Motivation

The purpose of this chapter is therefore twofold. First, to develop the theoretical framework needed to understand compressible inviscid flow at a level sufficient for engineering analysis. Second, to demonstrate, using a representative wedge-flow problem, how a modern density-based CFD solver can reproduce the classical analytical shock relations to engineering accuracy, and how the engineer must select solver settings, boundary conditions and mesh strategies that are appropriate to the physics being captured.

1.2 Information Propagation and Mach Number

When a body travels at $M < 1$, the pressure waves it generates outrun the body itself and “warn” the upstream fluid of its impending arrival. Streamlines have time to deflect smoothly and the flow adjusts in a continuous, nearly isentropic fashion. When the body exceeds the speed of sound ($M > 1$), the situation changes qualitatively. The body now outruns its own pressure signals; the fluid ahead is genuinely ‘blind’ to the approaching body. The pressure waves coalesce on a conical envelope known as the Mach cone, and where this coalescence is forced by a solid surface, an extremely thin region of intense compression appears: *the shock wave*. The present chapter focuses on the supersonic regime, $M_\infty \in [2, 4]$, which contains the essential physics of shocks and expansions while remaining safely below the hypersonic regime where the perfect gas assumption begins to break down.

1.3 From Navier–Stokes to Euler Equations

A real fluid is governed by the full compressible Navier–Stokes equations, which express conservation of mass, momentum and energy together with the constitutive law for viscous stresses and Fourier’s law for heat conduction. These equations are formidable, and solving them in three dimensions over a complete aircraft geometry is one of the grand challenges of computational science.

A great deal can, however, be learnt by deliberately discarding viscosity. In the *Euler equations*, all viscous stress terms and the thermal conduction term are removed:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0, \quad (1.1)$$

$$\frac{\partial(\rho \mathbf{V})}{\partial t} + \nabla \cdot (\rho \mathbf{V} \otimes \mathbf{V}) = -\nabla p, \quad (1.2)$$

$$\frac{\partial}{\partial t} [\rho (e + \tfrac{1}{2} V^2)] + \nabla \cdot [\rho (e + \tfrac{1}{2} V^2) \mathbf{V}] = -\nabla \cdot (p \mathbf{V}). \quad (1.3)$$

The justification for this drastic simplification rests on the structure of high-Reynolds-number flow. Outside the boundary layer, viscous stresses are extremely small and exert essentially no influence on the pressure field; the static pressure inside a thin attached boundary layer is, to leading order, equal to the inviscid pressure at its edge ($\partial p / \partial n \approx 0$). Consequently, the inviscid pressure distribution provides an excellent prediction of lift and of pressure-induced (wave) drag at small angles of attack and high Mach numbers.

The Euler model captures *exactly* those phenomena that are unique to compressible flow: shock waves, expansion fans, and wave drag. What it cannot capture is equally important to acknowledge: skin-friction drag, viscous boundary layers, flow separation, and the all-important shock-wave/boundary-layer interaction in which a shock impinging on a viscous layer can trigger separation and large-scale unsteadiness. For preliminary supersonic design, however, the Euler equations strike an excellent balance between physical fidelity and computational cost, and they form the basis for the simulations presented in this chapter.

1.4 Shock Wave Theory

1.4.1 Governing Equations Across a Discontinuity

Consider a control volume wrapped tightly around a stationary, planar normal shock wave. The flow enters region 1 with properties (u_1, p_1, ρ_1, T_1) and leaves region 2 with (u_2, p_2, ρ_2, T_2) .

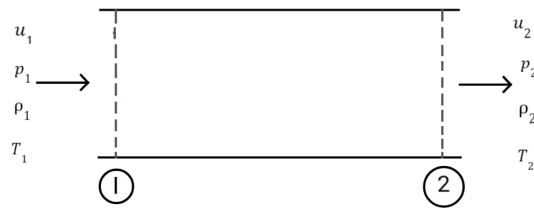


Figure 1.1: 1-D flow

The shock thickness is of the order of a few molecular mean free paths and is treated mathematically as a discontinuity. For steady, one-dimensional, adiabatic flow without body forces, the integral conservation laws reduce to:

1. **Continuity:** $\rho_1 u_1 = \rho_2 u_2$
2. **Momentum:** $p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$
3. **Energy:** $h_1 + \frac{1}{2} u_1^2 = h_2 + \frac{1}{2} u_2^2 = h_0$

Despite being adiabatic, shock flow is not isentropic. Entropy develops due to steep gradients reactivating molecular viscosity and thermal conduction in the shock layer. The shock lowers the stagnation pressure (p_0) while maintaining the stagnation temperature (T_0).

1.4.2 Pressure Ratio

To find the pressure ratio p_2/p_1 , we combine the continuity and momentum equations. From continuity, $u_2 = u_1(\rho_1/\rho_2)$. Substituting into the momentum balance and rearranging yields

$$p_2 - p_1 = \rho_1 u_1^2 \left(1 - \frac{\rho_1}{\rho_2} \right). \quad (1.4)$$

Introducing the freestream Mach number through $u_1^2 = M_1^2 a_1^2 = M_1^2 \gamma p_1 / \rho_1$ gives

$$\frac{p_2}{p_1} = 1 + \gamma M_1^2 \left(1 - \frac{u_2}{u_1} \right). \quad (1.5)$$

After invoking the energy equation and eliminating the velocity ratio, one arrives at the classical **Rankine–Hugoniot** relation:

$$\boxed{\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1} (M_1^2 - 1)} \quad (1.6)$$

Analogous derivations yield the temperature, density and downstream Mach number ratios:

$$\frac{T_2}{T_1} = \frac{[2\gamma M_1^2 - (\gamma - 1)][(\gamma - 1)M_1^2 + 2]}{(\gamma + 1)^2 M_1^2}, \quad (1.7)$$

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1)M_1^2}{(\gamma - 1)M_1^2 + 2}, \quad M_2^2 = \frac{(\gamma - 1)M_1^2 + 2}{2\gamma M_1^2 - (\gamma - 1)}. \quad (1.8)$$

For an oblique shock making an angle β with the freestream, the same relations apply provided M_1 is replaced by its component normal to the shock, $M_{n,1} = M_1 \sin \beta$. The deflection angle δ produced by the shock is then linked to β and M_1 by the θ – β – M relation.

1.4.3 Physical Interpretation: Why the Pressure Rises

Across a shock, the directed kinetic energy of the incoming supersonic stream is rapidly converted into random thermal motion of the molecules. The flow decelerates, the temperature rises, the density increases, and, as a direct consequence of the rise in both ρ and T , the static pressure jumps. The process is irreversible: entropy increases, and a portion of the available stagnation pressure is permanently lost. This is the fundamental origin of *wave drag* in supersonic flight.

1.4.4 Weak and Strong Shock Solutions

For a given upstream Mach number and deflection angle (below the maximum allowable value), the θ - β - M relation admits two solutions: a *weak* shock with smaller β and supersonic post-shock flow and a *strong* shock with larger β and subsonic post-shock flow. In essentially all external aerodynamic applications, where the downstream pressure is not artificially raised, nature selects the weak solution.

1.5 Expansion Waves and Isentropic Turning

The mirror image of an oblique shock occurs when a supersonic flow turns *away* from itself around a convex corner. Rather than coalescing into a single discontinuity, the disturbances spread out into a continuous fan of infinitesimally weak Mach waves – the *Prandtl-Meyer expansion fan*. Across this fan, the static pressure, static temperature and density decrease; the Mach number increases and the entropy is conserved.

A shock is irreversible and dissipative, whereas an expansion fan is isentropic and reversible. Supersonic flow is unique for its asymmetry between compression and expansion: nature compresses abruptly but expands continuously without loss. On the lower wedge surface of the test geometry studied in this chapter, the supersonic stream compresses through an oblique shock at the leading deflection and expands as it leaves the deflected surface, allowing for a clean simulation of both phenomena.

1.6 Attached and Detached Shock Waves

For a wedge of half-angle δ in a supersonic stream of Mach number M_1 , the θ - β - M relation possesses a real solution only if δ does not exceed a maximum value $\delta_{\max}(M_1)$. When $\delta < \delta_{\max}$, an oblique shock attaches to the leading edge of the wedge. When $\delta > \delta_{\max}$, no attached oblique solution exists, the shock physically detaches from the body and stands off ahead of it as a curved bow shock.

The detached configuration is qualitatively different. A small region near the centreline of the bow shock behaves almost like a normal shock, producing locally subsonic flow behind it and a much larger entropy jump than the equivalent attached oblique shock. The standoff distance, and hence the resulting wave drag, increases with deflection angle and decreases with Mach number. Blunt bodies, which always create disconnected shocks, are utilised at hypersonic speeds to reduce aerodynamic heating by pushing high-temperature gas away from the body. At supersonic speeds, sharp leading edges with associated shocks reduce wave drag and are desirable. Both regimes were captured in the present study by varying the deflection angle of the lower surface across and beyond $\delta_{\max}$.

1.7 Numerical Implementation: The Fluent Workflow

1.7.1 Computational Domain

The geometry was constructed in ANSYS DesignModeler as a rectangular two-dimensional domain with a deflection of length $L = 0.1$ m on the lower surface. A short straight section was placed upstream of the deflection so that the inflow boundary did not interfere with the leading-edge shock, and the domain was made tall enough that the oblique shock could intersect only the right-hand outflow boundary, never the upper boundary. Both the deflection angle δ and the freestream Mach number M_∞ were exposed as design parameters so that parametric studies could be run without rebuilding the geometry.

1.7.2 Boundary Conditions and Their Justification

Because the flow is everywhere supersonic outside the immediate vicinity of the wedge, the choice of boundary conditions is dictated by the direction in which information can propagate. The following selections were made:

- **Left side (inflow):** *Pressure far-field*. This is the canonical Fluent boundary for compressible external flows. It allows the user to specify the freestream Mach number, static pressure, temperature and flow direction simultaneously, and is mathematically consistent with the characteristic structure of the Euler equations at supersonic inflow.
- **Top side:** *Pressure far-field* as well. The use of a non-reflecting characteristic-based boundary here is essential to prevent spurious reflection of the oblique shock back into the domain.
- **Right side (outflow):** *Pressure outlet*. At supersonic outflow, all characteristics leave the domain and the imposed pressure value is, in principle, ignored; this is the desired behaviour.
- **Lower surfaces (Bottom 1, Deflection, Bottom 2):** *Wall* with the inviscid (slip) condition. Since no boundary layer exists in the Euler model, the wall enforces only flow tangency.

A symmetry boundary was deliberately *not* used on the upper surface, since the flow physics above and below the wedge are asymmetric. All specified pressures were interpreted as gauge pressures relative to an operating pressure of zero, which is the recommended practice for high-Mach-number simulations to avoid loss of numerical precision.

1.7.3 Mesh and Mesh-Independence Study

An unstructured triangular mesh was generated in ANSYS Meshing with local refinement near the deflection corner, where the strongest gradients are expected. To establish mesh independence, three meshes of approximately 20 000, 42 000 and 90 000 cells were tested at the baseline condition $M_\infty = 2$, $\delta = 15^\circ$. The extracted pressure ratio p_2/p_1 varied by less than 0.8% between the medium and fine meshes, while the shock thickness (measured as the distance over which p rose from 5% to 95% of its post-shock value) reduced from approximately ten cells on the coarse mesh to two cells on the fine mesh after gradient adaptation. The medium mesh, combined with two levels of pressure-gradient adaptation, was therefore selected as the production mesh for all subsequent simulations.

1.7.4 Solver Configuration and Numerical Schemes

The **density-based** solver was selected because it solves the continuity, momentum and energy equations in a coupled manner, which is essential for compressible flow where pressure, density and temperature are intimately linked through the equation of state. The pressure-based solver, designed primarily for low-Mach-number flow, would suffer from severe under-relaxation issues at $M > 1$.

- **Models:** The energy equation was enabled (it is automatically activated when the ideal-gas law is selected as the density model) and the viscous model was set to *Inviscid*, reducing the system to the Euler equations.
- **Materials:** Air was modelled as an ideal gas with $\gamma = 1.4$ and $R = 287 \text{ J}/(\text{kg}\cdot\text{K})$, with constant specific heats (calorically perfect gas).
- **Spatial discretisation:** A second-order upwind scheme was used for flow variables, with the AUSM (Advection Upstream Splitting Method) flux scheme. AUSM was chosen over the default Roe-FDS scheme because it provides sharper resolution of strong shocks while being substantially less prone to the so-called “carbuncle” numerical artefact that can corrupt bow shocks in particular.
- **Initialisation:** The flow field was initialised with freestream values computed from the inflow boundary, which provides a clean starting point for supersonic simulations.
- **Convergence:** The Courant number was started at 1 and gradually ramped up to 5 as the residuals dropped. Residuals for continuity, momentum and energy were monitored together with the integrated mass flux at the outflow; the simulation was deemed converged when residuals fell at least four orders of magnitude and the mass-flux imbalance was below 0.01%.

1.7.5 Extraction of Shock Ratios – The Plateau Method

Because the shock ratios p_2/p_1 , T_2/T_1 and M_2/M_1 are not native output quantities in Fluent, a consistent and reproducible extraction procedure was developed. A line probe was placed parallel to the freestream at a height of 50% of the domain, intersecting both the undisturbed upstream region and the post-shock region. Static pressure, temperature and Mach number were sampled along this line. The upstream values were taken as the spatial average over the well-defined plateau ahead of the shock, and the downstream values as the corresponding plateau behind it, deliberately avoiding the few cells inside the smeared shock layer where the gradients are still being captured. This “plateau method” is robust, physically meaningful and reproducible by any other user given the same mesh.

1.8 Results and Analysis

Simulations were carried out for $M_\infty = 2, 3$ and 4 at deflection angles ranging from 5° to 25° , covering both attached and detached shock regimes.

1.8.1 Shock Angle Behaviour

The first observation, evident already in the contour plots of Mach number, is that the shock angle β decreases monotonically with increasing M_∞ for a fixed deflection angle. At $M_\infty = 2$ and $\delta = 20^\circ$, the shock stands at $\beta \approx 53^\circ$, well separated from the wedge surface. At $M_\infty = 4$ and the same deflection, β has shrunk to roughly 32° , and the post-shock layer between the shock and the wall is markedly thinner. This compression of the shock layer is a direct precursor of hypersonic flow, where the shock eventually lies almost on top of the body surface. Physically, it reflects the fact that as M_∞ increases, the normal Mach number $M_1 \sin \beta$ required to support a given deflection is achieved at progressively smaller β .

1.8.2 Shock Wave Angle and Density Ratio Validation

Having established the numerical framework, we now turn to validating the solver’s ability to reproduce the fundamental relationships of oblique shock theory. We begin with the geometric characteristics of the shock wave itself.

The θ – β –M Relationship

Figure 1.2 presents the classical θ – β –M diagram, which relates the shock wave angle β to the flow deflection angle θ for a given freestream Mach number. This diagram is one of the most important tools in supersonic aerodynamics, as it immediately reveals whether

an attached oblique shock is physically possible for a given combination of M_∞ and wedge angle.

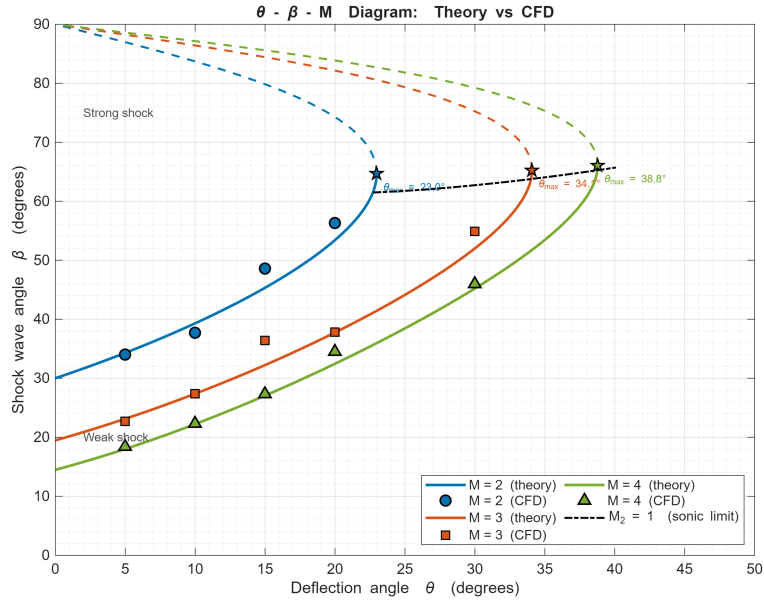


Figure 1.2: Shock wave angle β versus deflection angle θ for $M_\infty = 2, 3$, and 4 . Solid lines represent analytical oblique shock theory; markers denote CFD results. The dashed curves indicate the strong-shock branch, which is not realized in external flow configurations. Stars mark the maximum deflection angle $\theta_{\max}$ for each Mach number, beyond which the shock detaches.

The solid curves in Figure 1.2 are computed from the analytical θ - β - M relation, which may be derived by combining the tangency condition at the wedge surface with the normal-shock jump conditions. For each Mach number, two solution branches exist: the weak shock (lower β , supersonic post-shock flow) and the strong shock (higher β , subsonic post-shock flow), shown as dashed lines. In external aerodynamic flows where the downstream pressure is not artificially constrained, nature invariably selects the weak-shock solution.

Several fundamental features of supersonic flow are embodied in this diagram:

- 1. Mach number dependence.** For a fixed deflection angle, the shock angle β decreases as M_∞ increases. At $\theta = 20^\circ$, for instance, $M_\infty = 2$ produces $\beta \approx 53^\circ$, whereas $M_\infty = 4$ yields $\beta \approx 32^\circ$. This compression of the shock layer with increasing Mach number is a harbinger of hypersonic flow, where the shock eventually lies nearly flush with the body surface.
- 2. Maximum deflection angle.** Each Mach number admits a maximum deflection angle $\theta_{\max}$ beyond which no real solution exists for an attached oblique shock. This critical angle, marked by stars in Figure 1.2, increases with Mach number: $\theta_{\max} \approx 23^\circ$ at $M_\infty = 2$, rising to approximately 34° at $M_\infty = 3$ and 39° at $M_\infty = 4$.

3. The sonic boundary. The dashed horizontal line at $M_2 = 1$ represents the locus of points where the post-shock flow is exactly sonic. This line separates the weak-shock branch ($M_2 > 1$) from the strong-shock branch ($M_2 < 1$). All physically realized attached shocks in the present study lie on the weak branch, with supersonic downstream flow.

The discrete markers in Figure 1.2 represent shock angles extracted from the present Fluent simulations. The shock angle was determined by identifying the locus of maximum pressure gradient in the flow field and fitting a straight line through this region in the vicinity of the wedge leading edge. The agreement between CFD and theory is excellent across the entire weak-shock regime, with maximum deviations of less than 2° which is well within the uncertainty introduced by the finite thickness of the numerically captured shock and the discretization of the mesh. This close correspondence confirms that the inviscid Euler model, when properly discretized, faithfully reproduces the kinematic structure of oblique shocks.

Density Jump Across the Shock

While the shock angle β describes the geometry of the discontinuity, the density ratio ρ_2/ρ_1 quantifies the strength of the compression. Figure 1.3 presents this ratio as a function of deflection angle for the three Mach numbers studied.

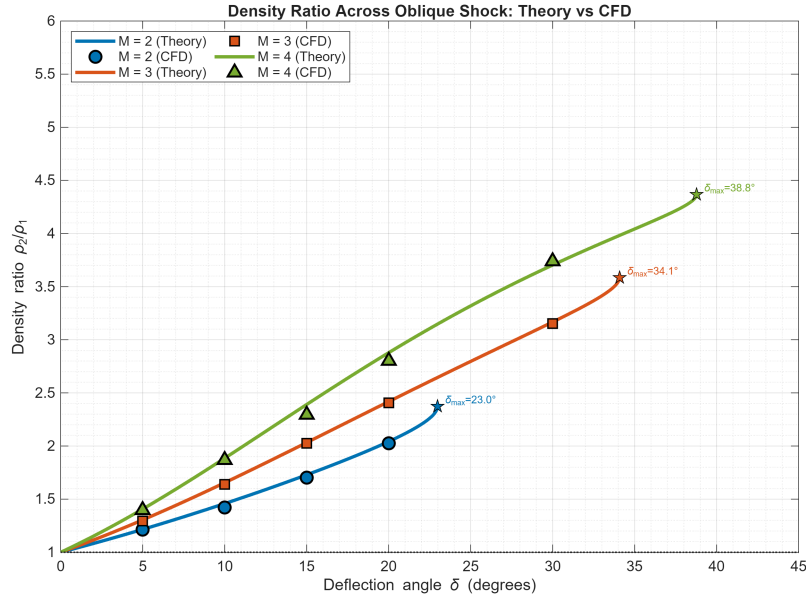


Figure 1.3: Density ratio ρ_2/ρ_1 across the oblique shock versus deflection angle θ for $M_\infty = 2, 3$, and 4. Solid lines represent the analytical Rankine–Hugoniot prediction; markers denote CFD-extracted plateau values. Stars indicate the maximum deflection angle for attached shocks. The density ratio increases nonlinearly with both deflection angle and freestream Mach number.

The theoretical curves are computed from the Rankine–Hugoniot density-jump relation, Eq. (1.8), with M_1 replaced by the normal component $M_{n,1} = M_\infty \sin \beta$, where $\beta(\theta, M_\infty)$ is obtained from the θ – β – M relation. The density ratio exhibits two important trends:

Nonlinear growth with deflection angle.

Even at moderate deflection angles, the density jump is substantial. At $M_\infty = 2$ and $\theta = 20^\circ$, the density increases by a factor of approximately 2.4 across the shock. This is not a small perturbation; the fluid is genuinely being compressed into roughly 40% of its original volume (per unit mass), with corresponding increases in pressure and temperature.

Strong Mach number sensitivity.

The density ratio increases rapidly with freestream Mach number. At the highest simulated condition - $M_\infty = 4$, $\theta = 39^\circ$ -the density ratio approaches 4.4, meaning the post-shock density is more than four times the freestream value. In the limit of very strong shocks ($M_{n,1} \rightarrow \infty$), the density ratio is bounded by $(\gamma + 1)/(\gamma - 1) = 6$ for a diatomic gas, but practical supersonic vehicles operate well below this theoretical maximum.

The CFD-extracted values, shown as discrete markers, were obtained using the plateau method described in Section 1.7.5. Overall, the attached-shock regime shows good agreement with theory, with relative errors below 3%. The stars indicating θ_{max} match the transition points in Figure 1.2, proving the dataset’s internal consistency. Note that the density jump is not regulated by isentropic relations. While the flow upstream and downstream of the shock can be approximated as isentropic outside the boundary layer, the shock itself is irreversible. The density ratio is governed by the balance of mass, momentum, and energy across the discontinuity, not by a polytropic process. Entropy is formed and stagnation pressure is lost. Shock waves cause drag in inviscid fluids due to the loss of stagnation pressure, which permanently reduces the flow’s usable work capacity. The successful validation of shock angle and density ratio confirms that the numerical technique accurately captures the physics of oblique shocks. We will analyse the thermodynamic jump conditions (pressure and temperature) that affect the structural and thermal stresses of supersonic vehicles.

1.8.3 Pressure and Temperature Ratios

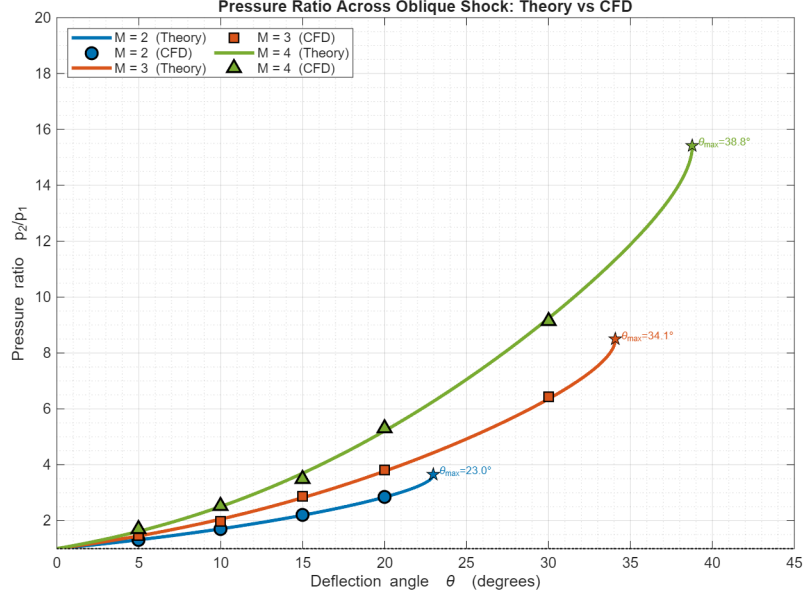


Figure 1.4: **Static pressure ratio p_2/p_1 across an oblique shock as a function of deflection angle θ for $M_\infty = 2, 3, 4$.** Solid lines correspond to the analytical Rankine–Hugoniot relations evaluated together with the θ – β – M relation, while circular markers denote the present Fluent CFD results extracted using the plateau method. Note the strongly nonlinear (approximately quadratic) growth of the pressure ratio with deflection angle, and its rapid increase with M_∞ . At $M_\infty = 4$ and $\theta = 20^\circ$ the pressure ratio exceeds 10, highlighting the extreme structural loading that supersonic vehicles must withstand.

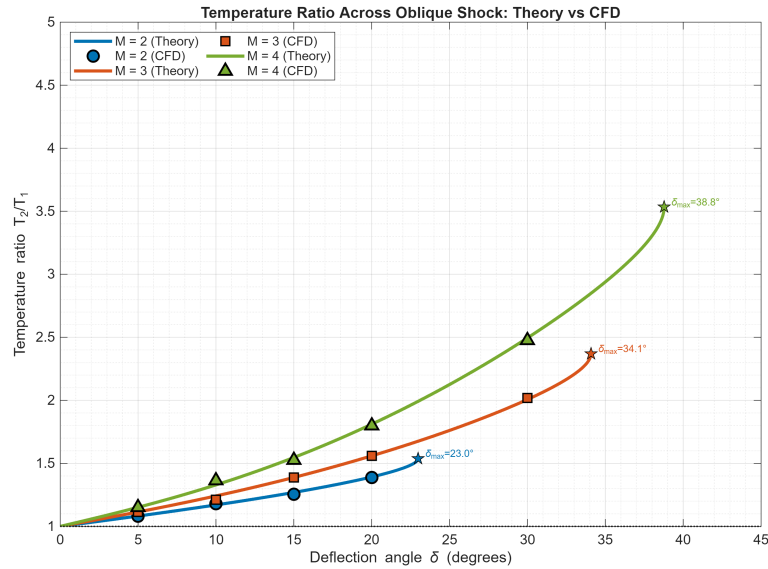


Figure 1.5: **Static temperature ratio T_2/T_1 across an oblique shock for $M_\infty = 2, 3, 4$.** The temperature rise is the direct fingerprint of the conversion of directed kinetic energy into random thermal energy across the shock. Even a moderate $M_\infty = 4$ shock with $\theta = 20^\circ$ approximately doubles the static temperature, which has obvious implications for thermal protection. The simulation faithfully reproduces the nonlinear analytical trend.

The agreement between the CFD data and the theoretical curves is excellent across the entire weak-shock branch, with the maximum relative error on the pressure ratio remaining below approximately 1.5% and that on the temperature ratio below 2%. The slight under-prediction observed at the largest deflection angles can be attributed primarily to numerical smearing of the shock over a finite number of cells, which biases the plateau-method downstream sampling toward slightly lower values. Mesh refinement reduces this error monotonically, in line with expectations for a second-order scheme.

1.8.4 Detached Shock Behaviour

At deflection angles beyond $\delta_{\max}(M_\infty)$, the simulations spontaneously produced a curved, detached bow shock standing ahead of the wedge. The stand-off distance grew with deflection angle and decreased with Mach number, reproducing the well-known qualitative trend. Behind the central, near-normal portion of the bow shock the flow was locally subsonic, with the corresponding Mach-number ratio dropping below unity – a regime that is impossible to reach with an attached oblique shock on the weak branch.

1.8.5 Mach Number Contours

The complete set of Mach number contour visualisations for all simulated configurations is presented in Figures 1.6 and 1.7.

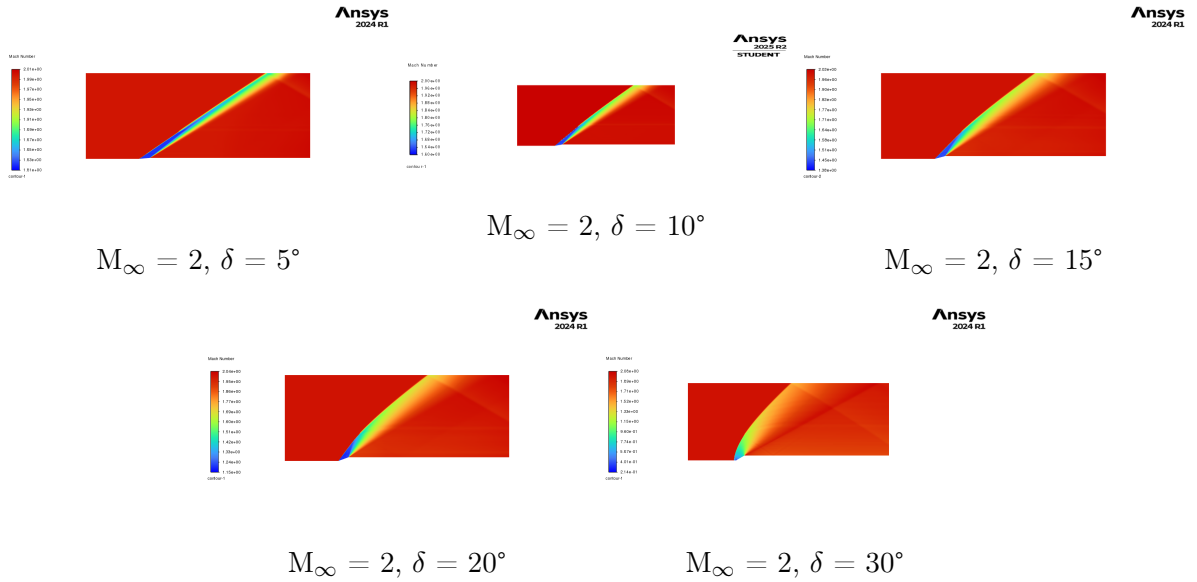


Figure 1.6: Mach number contours for various configurations showing the transition from attached oblique shocks to detached bow shocks.

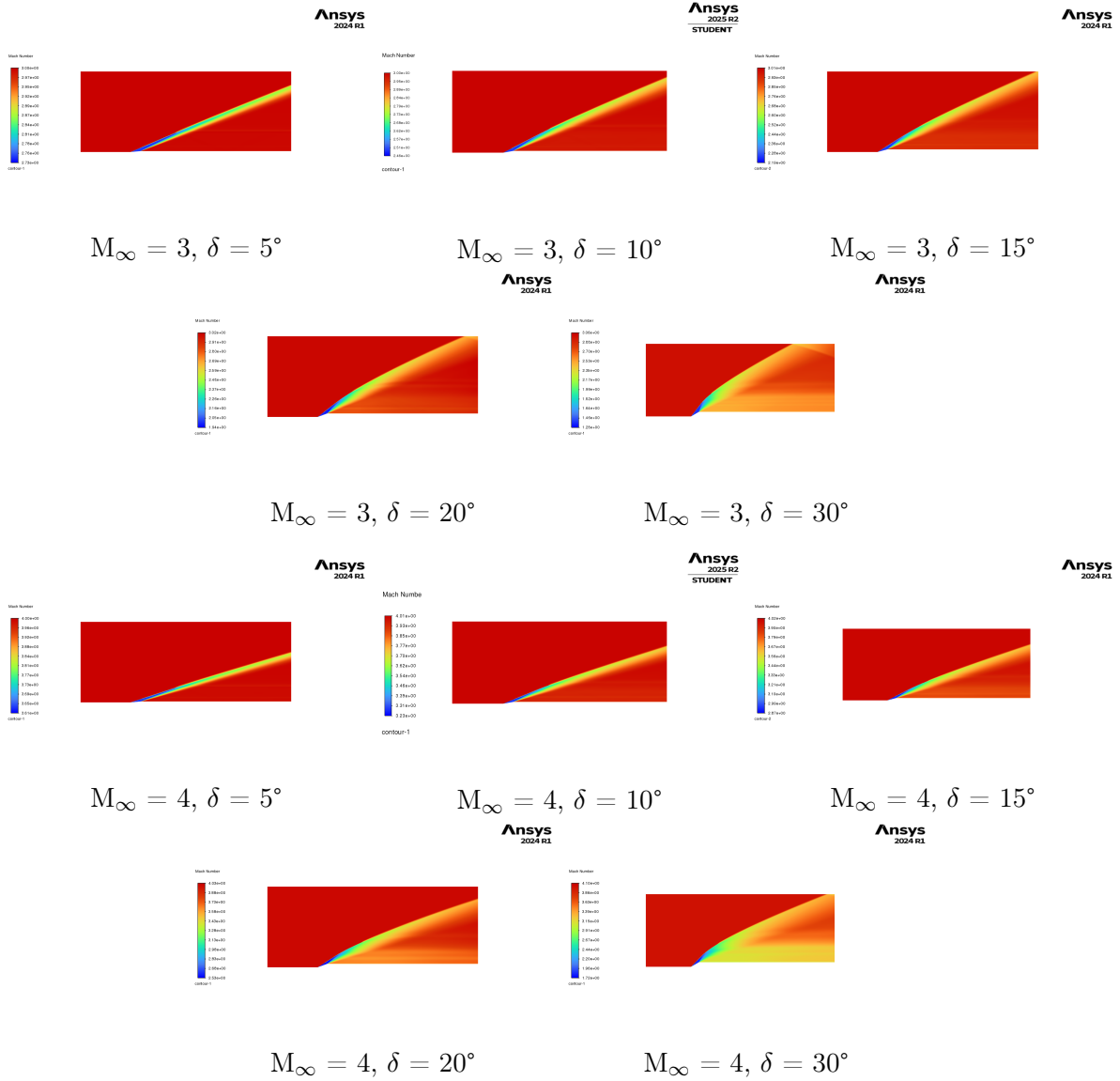


Figure 1.7: Mach number contours (continued) demonstrating the systematic decrease in shock angle and standoff distance with increasing Mach number.

1.9 Limitations of the Present Model

The good agreement between the Euler simulations and analytical theory should not be interpreted as a license to use this model for all supersonic problems. Several physical effects have been deliberately discarded:

- **No boundary layer.** The slip-wall condition removes viscous shear and skin-friction drag entirely. For a real wedge, a viscous boundary layer would form and could interact strongly with the leading-edge shock.
- **No shock–boundary-layer interaction.** The classical SWBLI phenomenon, in which a shock impinging on a viscous layer can trigger separation, recirculation and unsteadiness, is by definition absent from any inviscid model.

- **No viscous heating.** Although the shock temperature jump is captured, the additional aerodynamic heating produced by viscous dissipation in the boundary layer is not.
- **Calorically perfect gas.** At $M_\infty \gtrsim 5$, vibrational excitation, dissociation and ionisation of the air molecules render the constant- γ assumption invalid; the present model is therefore restricted to the supersonic, not the hypersonic, regime.
- **Two-dimensional idealisation.** Real wings have finite span and three-dimensional shock structures; the present analysis applies strictly to nominally 2D wedge flow.

Despite these limitations, the Euler approach remains an extremely valuable tool for preliminary design, parametric studies, and the rapid generation of initial conditions for higher-fidelity Navier–Stokes simulations.

1.10 Summary and Critical Assessment

This chapter has bridged the gap between the analytical theory of compressible flow and its numerical realisation in a modern CFD code. We began with the physical origin of the speed of sound and the qualitative distinction between subsonic and supersonic information propagation, and moved on to a full derivation of the normal-shock Rankine–Hugoniot relations. The complementary phenomenon of expansion waves was discussed, as was the transition between attached oblique and detached bow shocks as the deflection angle approaches its maximum allowable value.

On the numerical side, the choice of the density-based solver, the inviscid model, the AUSM flux scheme and pressure-gradient mesh adaptation was justified from first principles. A reproducible plateau-based extraction method was introduced for the post-shock quantities, and the results for $M_\infty = 2, 3$ and 4 were shown to match analytical theory within approximately 1.5% for the pressure ratio and 2% for the temperature ratio across the entire weak-shock branch.

Two broader engineering messages emerge from this work. First, theory and CFD are not competitors but partners: the analytical relations provide the verification benchmark that allows us to trust the solver, while the solver provides the geometric flexibility that allows us to attack problems that lie far beyond the reach of pencil-and-paper analysis. Second, the engineer must always be aware of *which* physics has been retained and which has been discarded. The Euler model used here is genuinely excellent at predicting shock locations, wave drag and post-shock temperatures, but it is silent on skin friction, separation and shock-boundary-layer interaction. Knowing the model’s domain of validity is, ultimately, just as important as knowing how to run it.